

Amplifier Review

What's an amplifier?

An amplifier is the most basic building block of almost any useful analog circuit (once we get past the basic resistors and capacitors). Schematically, a simple amplifier is shown below.

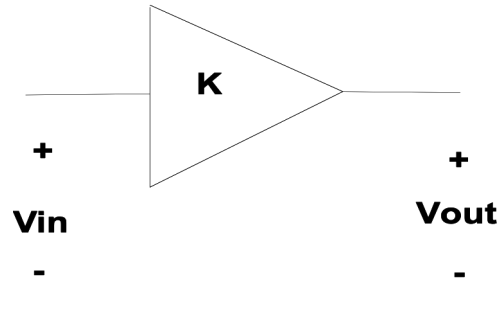


Figure 1: An amplifier

The figure shows a single-input amplifier. Simply, this amplifier multiplies its input by a number, called the *gain* of the amplifier, symbolized by K. Mathematically:

$$V_{\text{out}} = K \cdot V_{\text{in}}$$

Ideally, this gain should be large and easily adjustable, say, with a resistor.

Input and output resistance

As amplifiers are made out of common components like resistors, capacitors and transistors, they can be broken down into these parts for analysis. Two common parameters of interest are the *input* and *output* resistance (technically speaking that should be impedance, but we all know that) of an amplifier. For example, the input stage of an amplifier connected to a voltage source can be modeled as in Figure 2.

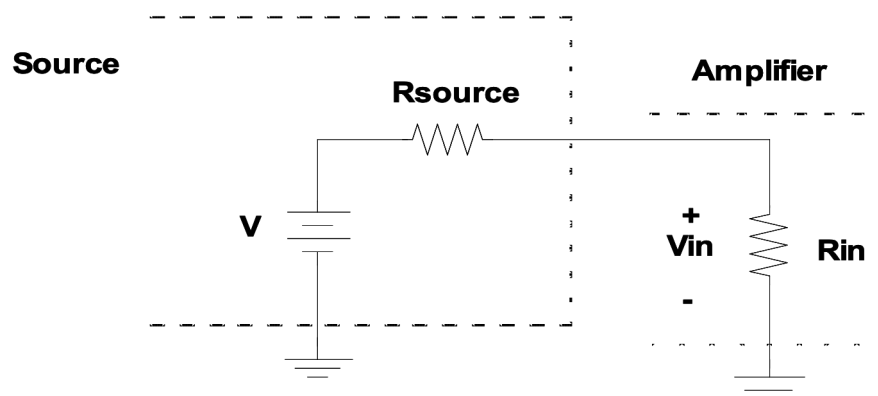


Figure 2: Amplifier input stage model

Both the source and input to the amplifier have impedances associated with them, as shown

above. So, the actual voltage seen by the amplifier is:

$$V_{input} = V \frac{R_{input}}{R_{input} + R_{source}}$$

Since we have no control over what sources may be connected to the amplifier, in order to measure as much of the source voltage as possible, the input resistance to the amplifier should be very high (ideally, infinite).

Similarly, one can show that the output resistance of an amplifier should be very low (ideally, 0) in order to transfer as much of the output of the amplifier as possible to its load.

Differential amplifiers

While the basic amplifier is a useful device, it suffers from some limitations. Chief among these, especially in the instrumentation arena, is noise. That is, whatever noise is present at the input to the amplifier gets amplified along with the desired signal. This leads directly to the differential amplifier, shown schematically below.

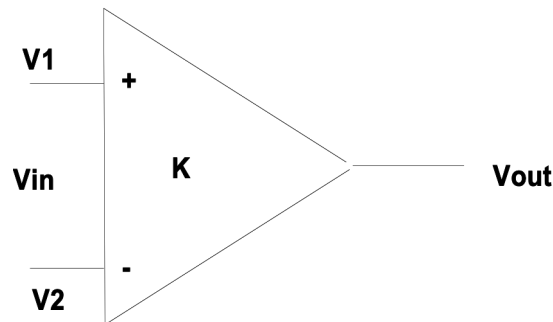


Figure 3: Differential amplifier

The immediately obvious difference between the single-ended amplifier above and the differential amplifier is that the differential amp has two inputs, marked with positive and negative signs (referred to as the *non-inverting* and *inverting* inputs, respectively). The output of this differential amp is given by:

$$V_{out} = K \cdot V_{in} = K \cdot (V_1 - V_2)$$

Thus, one can see the reason for the name: the differential amplifier amplifies only the difference between the two inputs. For our purposes, the most important consequence of this is noise reduction. Since the two input sources (perhaps a pair of electrodes on the skin) will likely be fairly close in all our applications, the noise that they see (e.g., interference from fluorescent lights) will be very similar if not the same (“if there’s any justice in the world”). Hence, this noise will not be amplified, only the signal in which we are interested.

Note that this is the real world with which we are dealing, and not circuit diagrams. No real differential amplifier can do this perfectly. Thus, an often-quoted specification of amplifiers is the common-mode rejection ratio (CMRR), which describes how well the amplifier does not amplify signals that are common to both inputs. One can determine this by taking a

ratio of the output of the amplifier with only one input driven to that with both inputs driven by the same source. Note that since this is given in units of V/V (and should be very large), CMRR is usually expressed in decibels (i.e., $20 \log_{10}$).

The operational amplifier: a special differential amplifier

Figure 3 is the symbol also used to indicate an operational amplifier (op amp), which is the most-commonly found differential amplifier. The ideal op amp has several characteristics that make it a desirable and useful device:

- Infinite input resistance
- Zero output resistance
- Infinite CMRR
- Infinite open-loop gain
- Easily selectable closed-loop gain

Note that, living in the real world, the word “infinite” should be replaced by the phrase “very high”, and “zero” by “low” in practice. Other things to keep in mind:

- All these parameters are frequency-dependent (yes, “gain-bandwidth product” is a real thing).
- Op amps require positive and negative power supplies, and the output will saturate as it approaches the supply voltages.
- Op amps are manufactured as integrated circuits, and thus typically are not able to generate large amounts of current (so you wouldn't want to drive speakers with them if you actually enjoy music).

We have already seen why high input resistance and CMRR and low output resistance are desirable. Open-loop gain refers to the gain of the op amp when there is no feedback from output to input – this configuration is not used when one uses the op amp as an amplifier, as a very small input voltage difference can produce a very large output (it can make for a quick and dirty comparator circuit). Closed-loop gain refers to the gain of the amplifier in one of several configurations where the output is fed back to the input as shown in the circuits below.

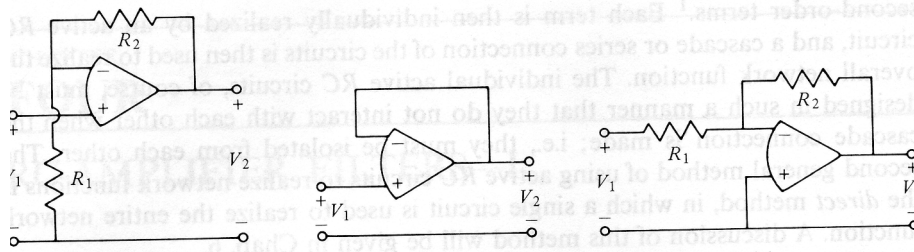


Figure 4: Popular opamp circuits; non-inverting amplifier, unity gain buffer, inverting amplifier.

One can easily derive the gains of these circuits using two simple rules for op amps:

1. The two inputs remain at the same voltage (virtual short circuit between input terminals)
2. No current flows into the op amp inputs

The gains are, respectively:

$$K_{\text{non-inverting}} = 1 + R_2/R_1$$

$$K_{\text{buffer}} = 1$$

$$K_{\text{inverting}} = -R_2/R_1$$

which are independent of the op amp. Other feedback configurations are also used; what they have in common is that the gain of the circuit is dictated not by the amplifier, but by the resistors being used (and resistors can be manufactured very accurately and cheaply).

Be aware that not pictured in any of these diagrams are the other connections found on a typical op amp IC package – power connections (positive and negative supplies), and offset and trim voltages.